SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	R. V. Harsha Vardhan	**Reg. No.:**	192211868
Section / Batch:	CSE	**Date:**	08/07/2026

Code of Conduct

I R.V. Harsha Vardhan/192211868 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ______________________

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20

GRAND TOTAL: **100 Marks**

Q1 **C Code – Pointer with Array** 10 marks

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Your Answer

30

Save Answer

Q2 **C Code - Linked List Node** 10 marks

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20

Q3 **C Code - Binary Search Complexity** 10 marks

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

O(log n)

Save Answer

Q4 **C Code - Pointer Increment** 10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10

Save Answer

Q5 **C Code - Array Address** 10 marks

```
int arr[]={5,10,15};
printf("%d",*(arr+1));
```

Your Answer

10

Save Answer

Q6 **C Code – Linked List Traversal** 10 marks

```
Node1 -> Node2 -> Node3 (Node values are 10,20,30)
current=head;
current=current->next;
printf("%d",current->data);
```

Your Answer

20

Save Answer

Q7 **C Code – Recursion (Factorial)**

10 marks

```
#include <stdio.h>
int fact(int n){
if(n==0)
return 1;
return n * fact(n-1);
}
int main(){
printf("%d", fact(4));
return 0;
}
```

Your Answer

24

CO1_AT1_C CODE OUTPUT PREDICTION

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QUESTIONS

Q1
C Code – Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q6

Q8 **C Code – Recursion (Sum)**

10 marks

```
#include <stdio.h>
int sum(int n){
if(n==0)
return 0;
return n + sum(n-1);
}
int main(){
printf("%d", sum(5));
}
```

Your Answer

15